Durham University

Examination Paper

Examination Session:	Year:	Exam Code:
May/June	2024	ENGI3391-WE01

Title: (BEng/MEng)

Control and Signal Processing 3

Time Allowed:	2 hours
Additional Material provided:	Yes – Formula Sheet included as part of exam paper. Graph paper provided.
Materials Permitted:	N/A
Calculators Permitted:	Those from the Casio fx-83 and fx-85 series.

Instructions to Candidates:	This is an in-person, invigilated exam. Answer ALL questions. Write your answers in the white-covered answer booklet with barcodes. Begin your answer to each question on a new page. All relevant workings must be shown.

Revision:	

Section A

Q.1

(a) A multi-path radio channel can be represented by the system in Figure Q.1 where a_k is the gain factor, T is time delay and $\delta(.)$ is the impulse function. Find the following:

(i) The impulse response and sketch it. [16%]
(ii) Deduce the unit step response. [10%]

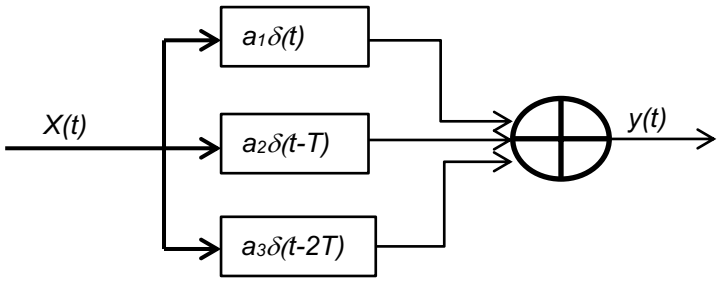


Figure Q.1

(b) Give four applications of spectrum analysis. [16%]
(c) Give the symmetry property of the double sided spectrum [8%]
(d) Find the magnitude and phase of the spectrum of the signal given by Eqn. 1 and sketch the single sided and double sided spectrum representation:

$$x(t) = 1 + sin\,\omega_o\,t + 2\,cos\,\omega_o\,t + 3\,cos\left(4\omega_o t + \frac{\pi}{8}\right) + 2\,sin\left(8\omega_o t + \frac{\pi}{4}\right) \qquad \text{Eqn. 1}$$

[25%]

(e) A power transmitter of a mobile phone operator is rated at 30 dBm. Give the transmitted power in W.

[10%]

(f) Derive an expression for the energy of a signal formed by multiplying a narrow pulse of duration T by a sinusoid $cos\omega_o t$ where $\omega_o = 2\pi k/T$ where k is an integer.

[15%]

Q.2

(a) A discrete casual time system has an impulse response given by

h[n]={1 5 6 8 9}

Find the response of the system to the following input signals

(i) An impulse $\delta[n]$. [6%]
(ii) A delayed impulse $\delta[n-2]$ [6%]
(iii) A sequence
x[n]={2 4 6 7} [18%]

(b) Find the Fourier transform of *g(a,t)* defined by Eqn. 2

$$g(a,t) = \begin{cases} \exp(-at) & t > 0 \\ 0 & t = 0 \\ -\exp(at) & t < 0 \end{cases}$$ Eqn. 2

[40%]

(c) Discuss the following

(i) Difference between a continuous and a discrete time signal [20%]
(ii) Analysis and synthesis of signals and systems [10%]

Section B

Question 3

(a) State the difference between practical phase-lead compensator design and practical phase-lag compensator design, including their design objectives, principles, and effects on the system performance.

[20%]

(b) With a gain controller, the system shown in Figure Q3.1 is always unstable. Explain the reason.

[20%]

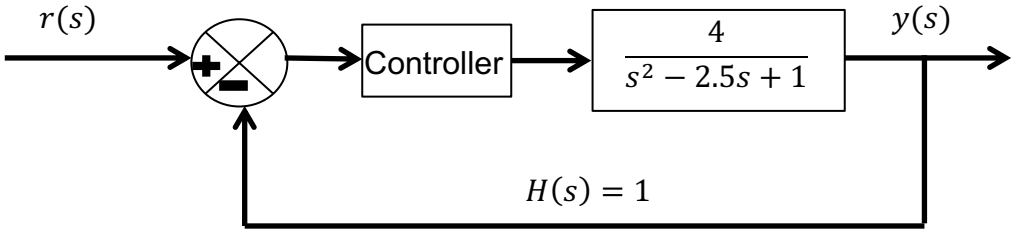


Figure Q3.1: A closed-loop control system with negative unit feedback

(c) If the controller in Figure Q3.1 is an ideal phase-lead compensator $k(s + 3)$, sketch the root locus of this closed-loop system by using the graph paper provided. Give the procedures and steps of the root locus plotting.

[20%]

(d) If the controller in Figure Q3.1 is an ideal phase-lead compensator $k(s + z)$, design this phase-lead compensator to meet the requirements of the damping ratio $\zeta = 0.4$ and the undamped natural frequency $\omega_n = 10$ rad/s.

[20%]

(e) Discretise your designed compensator in (d) by using backward difference rule, i.e., $z = \frac{1}{1-sT}$ where the sampling interval T is 0.2 second. Compare the forward difference and backward difference rules in terms of stability of the mapped system.

[20%]

Question 4

(a) State the advantages of digital compensators over analog compensators. State the challenging issues of designing a digital compensator?

[20%]

(b) The open-loop frequency response of a unity feedback control system is of the form:

$$GH(s) = \frac{K}{s(s+1)(s+2)}$$

When *K*=*2*, the open loop frequency response data is given in Table Q4 below.

i). Present this system's frequency response using a Nichols Chart.

[20%]

ii). State the meanings of gain margin and phase margin. Explain how you read them from a Nichols Chart.

[10%]

Table Q4

ω (rad/s)	Gain	Phase (degree)
0.2	4.88	-107
0.4	2.28	-123
0.6	1.37	-137
0.8	0.91	-150
1.0	0.63	-161.6
1.5	0.30	-183.2
2.0	0.16	-198.4
3.0	0.06	-217.9
4.0	0.03	-229.4

iii). Determine the phase margin and the gain crossover frequency.

[10%]

iv). Using this information in iii) to estimate the damping ratio and peak overshoot of the system's closed-loop step response.

[20%]

v). If you can adjust *K*, calculate the value of *K* that would provide a phase margin of 70 degrees. What is the critical gain for this system?

[20%]

Signal Processing Formula Sheet

Energy of a signal

$$E = \int_{-\infty}^{\infty} |f(t)|^2 dt$$

Power of a signal

$$P_{av} = \lim_{T\to\infty} \frac{1}{T} \int_{-T/2}^{T/2} |f(t)|^2 dt$$

Fourier series

$$x(t) = a_o + \sum_{k=1}^{\infty} \left[a_k \cos k\omega_o t + b_k \sin k\omega_o t \right]$$

where

$$a_o = \frac{1}{T} \int_T x(t)dt = dc\ term$$

$$a_k = \frac{2}{T} \int_T x(t) \cos k\omega_o t dt \qquad b_k = \frac{2}{T} \int_T x(t) \sin k\omega_o t dt$$

Alternative Fourier series

Trigonometric form:

$$x(t) = a_o + \sum_{k=1}^{\infty} \sqrt{a_k^2 + b_k^2} \cos(k\omega_o t + \phi_k)$$

where

$$\phi_k = -\tan^{-1} \frac{b_k}{a_k}$$

Complex exponential form

$$x(t) = \sum_{k=-\infty}^{\infty} d_k\, e^{jk\omega_o t}$$

where

$$\boldsymbol{d_k = \frac{1}{T} \int_T x(t).\, e^{-jk\omega_o t}\, dt}$$

$d_k = \frac{1}{2}(a_k - jb_k)\ for\ k > 0\ and\ d_k = \frac{1}{2}(a_k + jb_k)\ for\ k < 0$

Fourier transform

$$\mathcal{F}\{x(t)\} = X(\omega)$$
$$= \int_{-\infty}^{\infty} x(t)e^{-j\omega t}dt$$

Alternatively it can be written in terms of the frequency in cycles per seconds or Hz as

$$\boldsymbol{X(f) = \int_{-\infty}^{\infty} x(t)e^{-j2\pi ft}dt}$$

Inverse Fourier transform

$$x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(\omega)e^{j\omega t}d\omega$$

or

$$\boldsymbol{x(t) = \int_{-\infty}^{\infty} X(f)e^{j2\pi ft}df}$$

Fourier transform properties

1. **Linearity**

$x_1(\mathrm{t}) \leftrightarrow X_1(\omega)$, and $x_2(\mathrm{t}) \leftrightarrow X_2(\omega)$ then
$\mathrm{a}x_1(\mathrm{t}) + bx_2(t) \leftrightarrow aX_1(\omega) + bX_2(\omega)$

2. **Differentiation**

$$y(t) = \frac{d^n x(t)}{dt^n} \leftrightarrow (j\omega)^n X(\omega)$$

3. **Integration**

$$y(t) = \int_{-\infty}^{t} x(\tau)d\tau \leftrightarrow \frac{1}{j\omega}X(\omega) + \pi X(0)\delta(\omega)$$

4. **Time scaling**

$$x(b\,\mathrm{t}) \leftrightarrow \frac{1}{|b|}X\left(\frac{\omega}{b}\right)$$

CONTINUED

5. **Time reversal**

$$x(-\mathrm{t}) \leftrightarrow X(-\omega) = X^{*}(\omega)$$

6. **Time delay**

$$y(t) = x(\mathrm{t}\text{-}\mathrm{T}) \leftrightarrow X(\omega)\, \mathrm{e}^{-\mathrm{j}\omega \mathrm{T}}$$

7. **Multiplication in time**

$$x_1(\mathrm{t}).x_2(\mathrm{t}) \leftrightarrow \frac{X_1(\omega) * X_2(\omega)}{2\pi}$$

8. **Multiplication in frequency (convolution in time)**

$$X_1(\omega)X_2(\omega) \leftrightarrow x_1(\mathrm{t}) * x_2(\mathrm{t})$$

9. **Modulation property**

$$x(\mathrm{t})\cos(\omega_o t) \leftrightarrow \frac{1}{2}X(\omega - \omega_{\mathrm{o}}) + \frac{1}{2}X(\omega + \omega_o)$$

10. **Duality property**

If $x(t) \leftrightarrow X(\omega)$

Then $X(t) \leftrightarrow 2\pi x(-\omega)$

Discrete Fourier Transform

Discrete Fourier Series

DFS: $d_k = X[k] = \frac{1}{N}\sum_{n=0}^{N-1} x[n] e^{\frac{-j2\pi kn}{N}}$

Inverse Discrete Fourier Series

IDFS: $x[n] = \sum_{k=0}^{N-1} d_k e^{\frac{j2\pi kn}{N}}$

Euler's identities

$$A\sin(\omega_o t) = \frac{A}{j2}\left(e^{j\omega_o t} - e^{-j\omega_o t}\right)$$

$$A\cos(\omega_o t) = \frac{A}{2}\left(e^{j\omega_o t} + e^{-j\omega_o t}\right)$$

$$A\cos(\omega_o t + \phi_o) = \frac{A}{2}\left(e^{j\omega_o t}e^{j\phi_o} + e^{-j\omega_o t}e^{-j\phi_o}\right)$$

$e^{\pm j\theta} = \cos\theta \pm j\sin\theta$

Impulse function in continuous time domain

$\delta(t) = 0\ for\ t \neq 0\ and\ infinite\ at\ t = 0$

$$\int_{t=-\infty}^{\infty} \delta(t) = 1$$

Impulse function in discrete time domain

$\delta[n] = 1\ n = 0\ and\ 0\ elsewhere$

Unit step function in continuous time domain

$u(t) = 1\ for\ t \geq 0\ and\ 0\ t < 0$

Unit step function in discrete time domain

$u[n] = 1\ for\ n \geq 0\ and\ 0\ for\ n < 0$

Convolution integral

$$y(t) = x_1(t) * x_2(t) = \int_{-\infty}^{\infty} x_1(\tau)x_2(t-\tau)d\tau = \int_{-\infty}^{\infty} x_1(t-\tau)x_2(\tau)d\tau$$

Convolution sum

$$y[n] = \sum_{k=-\infty}^{\infty} x_1[k]x_2[n-k] = \sum_{k=-\infty}^{\infty} x_1[n-k]x_2[k]$$

Convolution with an impulse

$x(t) * \delta(t - t_o) = x(t - t_o)$

Correlation functions

Continuous time auto-correlation

Power signal: $\phi_{xx}(\tau) = \lim_{T\to\infty} \frac{1}{2T} \int_{-T}^{T} x(t)x(t+\tau)dt$

Energy signal: $\phi_{xx}(\tau) = \int_{-\infty}^{\infty} x(t)x(t+\tau)dt$

Continuous time cross-correlation:

Power signals: $\phi_{xy}(\tau) = \lim_{T\to\infty} \frac{1}{2T} \int_{-T}^{T} x(t)y(t+\tau)dt$

Energy signals: $\phi_{xy}(\tau) = \int_{-\infty}^{\infty} x(t)y(t+\tau)dt$

Discrete time auto-correlation:

Power signals: $\phi_{xx}[k] = \frac{1}{N} \sum_{n=0}^{N-1} x[n]x[n+k]$

Energy signals: $\phi_{xx}[k] = \sum_{n=0}^{N-1} x[n]x[n+k]$

Discrete time cross-correlation:

Power signals: $\phi_{xy}[k] = \frac{1}{N} \sum_{n=0}^{N-1} x[n]y[n+k]$

Energy signals: $\phi_{xy}[k] = \sum_{n=0}^{N-1} x[n]y[n+k]$

Fourier transform of an impulse

$\delta(t) \leftrightarrow 1$

Inverse Fourier transform of an impulse

$\delta(\omega) \leftrightarrow 1/2\pi$

Roots of a second order polynomial

$ax^2 + bx + c = 0$

are given by

$$x_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Trigonometric identities

$\cos\theta = \cos(-\theta) = \sin(\pi/2 - \theta)$

$\sin\theta = -\sin(-\theta) = \cos(\pi/2 - \theta)$

$\tan\theta = -\tan(-\theta) = \cot(\pi/2 - \theta)$

$\sin 2\theta = 2\cos\theta \sin\theta$

$\cos 2\theta = \cos^2\theta - \sin^2\theta = 2\cos^2\theta - 1 = 1 - 2\sin^2\theta$

$\tan 2\theta = 2\tan\theta / (1 - \tan 2\theta)$

$$\sin\alpha.\sin\beta = \frac{1}{2}\{(\cos(\alpha - \beta) - cos(\alpha + \beta)\}$$

$$\cos\alpha.\cos\beta = \frac{1}{2}\{(\cos(\alpha - \beta) + cos(\alpha + \beta)\}$$

$$\sin\alpha.\cos\beta = \frac{1}{2}\{(\sin(\alpha - \beta) + sin(\alpha + \beta)\}$$

$\sin\theta/2 = \pm\sqrt{[(1 - \cos\theta)/2]}$

$\cos\theta/2 = \pm\sqrt{[(1 + \cos\theta)/2]}$

$\tan\theta/2 = \sin\theta/(1 + \cos\theta)$

$\cos^2\theta + \sin^2\theta = 1$

$\sec^2\theta - \tan^2\theta = 1$

$\mathrm{cosec}^2\theta - \cot^2\theta = 1$

$\sin^2\theta = \frac{1}{2}(1 - \cos 2\theta)$

$\cos^2\theta = \frac{1}{2}(1 + \cos 2\theta)$

$\tan^2\theta = (1 - \cos 2\theta)/(1 + \cos 2\theta)$

$\sin(A + B) = \sin A \cos B + \cos A \sin B$

$\sin(A - B) = \sin A \cos B - \cos A \sin B$

$\cos(A + B) = \cos A \cos B - \sin A \sin B$

$\cos(A - B) = \cos A \cos B + \sin A \sin B$

$\tan(A + B) = (\tan A + \tan B)/(1 - \tan A \tan B)$

$\tan(A - B) = (\tan A - \tan B)/(1 + \tan A \tan B)$

$\sin A + \sin B = 2 \sin ½(A + B) \cos ½(A - B)$

$\sin A - \sin B = 2 \cos ½(A + B) \sin ½(A - B)$

$\cos A + \cos B = 2 \cos ½(A + B) \cos ½(A - B)$

$\cos A - \cos B = - 2 \sin ½(A + B) \sin ½(A - B)$

$\tan A + \tan B = \sin (A + B)/(\cos A \cos B)$

$\tan A - \tan B = \sin (A - B)/(\cos A \cos B)$

$\sin2A + \sin2B = 1 - \cos(A + B)\cos(A - B)$

$\sin2A - \sin2B = \sin(A + B)\sin(A - B)$

$\cos2A + \sin2B = 1 - \sin(A + B)\sin(A - B)$

$\cos2A - \sin2B = \cos(A + B)\cos(A - B)$

$\cos2A + \cos2B = 1 + \cos(A + B)\cos(A - B)$

$\cos2A - \cos2B = - \sin(A + B)\sin(A - B)$

For a triangle with sides a, b, c, and angles A, B, C opposite sides a, b and c respectively, the following relations hold.

$a2 = b2 + c2 - 2\ b\ c \cos A$

$a/\sin A = b/\sin B = c/\sin C.$

$(a - b)/(a + b) = \tan ½(A - B)/ \tan ½(A + B)$

L3 Control Formula Sheet

Standard second-order transfer function

$$\frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

Closed-loop poles of under-damped second-order system

$$s_{1,2} = -\zeta\omega_n \pm j\omega_n\sqrt{1-\zeta^2}$$

The overshoot percentage (%) of the standard second-order system

$$M_o = 100\exp\left(\frac{-\pi\zeta}{\sqrt{1-\zeta^2}}\right)$$

The settling time for a second-order system can be defined as

$$T_s = \frac{\text{Number of time constants}}{\zeta\omega_n}$$

The relation of number of time constants and settling error

Number of time constants	$e^{-\frac{t}{\tau}}$	Settling Error (%)
1	0.368	36.8
2	0.135	13.5
3	0.05	5
4	0.018	1.8
5	0.007	0.7

Defining $u(t)$ as the controller output and $e(t)$ as the error, the standard form of the PID algorithm is:

$$u(t) = K_p e(t) + K_i \int_0^t e(\tau)d\tau + K_d \frac{d}{dt}e(t)$$

Laplace form of the PID controller

$$G(s) = K_p + \frac{K_i}{s} + K_d s$$

7 Steps for Root Locus Sketching:

CONTINUED

1). Write the characteristic function so that *K* is a multiplier *1+KG(s)=0.*

2). Factor *G(s)* in terms of *n* poles and *m* zeros:

$$G(s) = \frac{\prod_{i=1}^{m}(s - z_i)}{\prod_{j=1}^{n}(s - p_j)}$$

3). Mark poles and zeros on the *s*-plane.

4). Determine the number of separate loci (lines)

5). Locate the segments of the real axis that are root loci

6). The remaining loci proceed to the zeros at infinity along asymptotes centered at σ and with angle *Φ*

$$\sigma = \frac{\sum_{i=1}^{n} p_i - \sum_{j=1}^{m} z_j}{n - m}$$

$$\phi = \frac{(2q+1)}{n-m}\pi,$$

where $q = 0,1,\ldots,(n-m-1)$

7). Plot remaining root locus.

Magnitude Criteria:

$$K = \frac{\prod_{i=1}^{n}|s - p_i|}{\prod_{j=1}^{m}|s - z_j|}$$

Angle Criteria:

$$\sum_{i=1}^{n}\phi_{p_i} - \sum_{j=1}^{n}\phi_{z_i} = (2q+1)\pi$$

END